

Erwhan Le Coueffec

Generalist R&D Engineer - Photonics & Optical Systems

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PROFILE

CentraleSupélec engineering student specializing in photonics, with hands-on experience in optical metrology, calibration, instrumentation and experimental design. Seeking an R&D position in Switzerland with a strong experimental component.

EXPERIENCE

Holo3 Oct. 2025 - Sep. 2026

R&D Engineer Apprentice

- Improved HoloScan calibration by comparing calibration strategies on identical acquisitions and validating with a certified 250.1097 mm ball bar; maximum error remained below 60 um at every tested position, versus roughly 120 um previously.
- Introduced a Design of Experiments approach with around 30 trials across 7-8 camera/lighting factors; used multiple linear regression and identified black/white target contrast as the dominant factor.
- Characterized an internal vision measurement machine for accuracy, repeatability and uncertainty.

Placoplatre Oct. 2023 - Jun. 2025

WCM / Continuous Improvement Engineer Apprentice

- Worked on autonomous maintenance, 5S, industrial risk identification and a visual OK/NOK support to simplify plaster-viscosity control.

Wilo France Sep. 2021 - Aug. 2022

Measurement / Industrialization Apprentice

- Reduced leak-test complexity for 16 HomeBooster pump models to 3 ATEQ test programs and configured the programs directly.
- Defined, measured and deployed a certified Electrostatic Protected Area (EPA) on the Plavis production line.

EDUCATION

CentraleSupélec - Metz 2023 - 2026

Generalist Engineering Program - Apprenticeship (FISA)

Third-year focus: Physics & Nanotechnology - Photonics and Nano-systems Engineering. Engineering degree requirements expected to be completed by January 2027.

Lycee Joliot-Curie - Rennes 2022 - 2023

Classe preparatoire ATS

Post-technical-degree preparation for engineering-school admission.

Universite du Mans 2020 - 2022

DUT Mesures Physiques

Second year completed as an apprentice at Wilo France; mixed focus on materials and measurement systems.

TECHNICAL SKILLS

Metrology / R&D	Optical metrology, instrumentation, experimental design, uncertainty analysis, industrial cameras
Optics / lab	Optical benches, lasers, objectives, polarizers, half/quarter-wave plates, spectrometer, oscilloscope, function generator
Data / software	MATLAB, advanced Excel (Solver, regression, formulas), LTspice, LabVIEW (previous experience), Python/VBA (basic), SolidWorks (previous experience)

LANGUAGES & INTERESTS

French - Native | English - B1/B2 | German - Beginner

Handball - 5 years, amateur/departmental level | Boulderling - 5C to 6A